LI, WENTAO

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EDUCATION

The University of Texas Health Science Center at Houston *Feb 2021 - Dec 2025*
4^{th}-year PhD student in the School of Biomedical Informatics
Awards: *Dean's Excellent Award 2021, 2021; Jingchun Sun Memorial Scholarship of UTHealth, 2023; D.Bradley McWilliams Scholars Endowed Scholarship Award, 2024.*

University of California, San Diego *Sep 2018 - Jun 2020*
Master of Science in Statistics

Shanghai Maritime University, Shanghai, China *Sep 2014 - Jun 2018*
Bachelor of Science in Mathematics
Awards: *Dean's List of SMU, 2016; First Class Scholarship of SMU, 2017.*

EXPERTISE HIGHLIGHTS

- Proficient in Git for version control and collaboration
- Experience in Docker for streamlined application deployment and environment management
- Strong coding skills with deep learning platforms (PyTorch and TensorFlow) in Python and R
  - Design and implement state-of-the-art deep learning algorithms, such as Transformer, Diffusion model, contrastive learning, etc.
  - Hands-on experience in statistics topics including data imputation, survival analysis, regression, etc.
- Proficient in Linux and working with HPC clusters and distributed computing environments
  - Developed Federated Learning frameworks that allow cross-silo training
  - Deployed Federated Learning network across Houston, San Diego, and Munich
- Comprehensive experience in Bioinformatics and medical imaging research
  - Build chest-CT Foundation model with CLIP and DINOv2
  - Analyze multi-omics data from public databases and UT MD Anderson in-house data
  - Develop cross-modality Genome-Wide Association Studies method

RESEARCH EXPERIENCE

The University of Texas MD Anderson Cancer Center *Oct 2023 - present*

· Design and deploy a foundation model for chest CT scans;
· Developed a novel cross-modal attention fusion method to enhance predictive modeling in cancer studies[1];
· Performed survival analysis on lung cancer patients in immunotherapy and chemotherapy[2];
· Investigated brain regional interactions and genetic variant expressions with a novel approach that incorporates individualized brain patterns in genomic association analysis.

The University of Texas Health Science Center at Houston *Jul 2020 - Sep 2023*

· Developed COLLAGENE, a secure analysis tool offers a practical solution for privacy-preserving GWAS[3];
· Developed series of Federated Generalized Linear Mixed Models (FedGLMMs) for Genome-Wide Association Studies (GWAS)[4, 5];
· Engineered privacy-preserving correlation estimation and genetic imputation algorithms for GWAS, enhancing data privacy without compromising analytical power;
· Developed a privacy-preserving federated learning method for cohort studies[6];
· Created a deep learning model to accurately predict blood pressure from photoplethysmogram (PPG) signals;
· Hosted federated training across Houston, San Diego, and Munich with VERTIGO-CI[7];

University of California, San Diego *Jun 2019 - Jun 2020*

- Conducted mathematical proofs for calibration measurements and models in clinical prediction research [8];

CONFERENCE PRESENTATIONS

AMIA 2021 Virtual Informatics Summit *Mar 2021*
Principal Speaker

- Presentation on published conference paper *VERTIcal Grid lOgistic regression with Confidence Interval.*

MICCAI 2024 CMMCA Workshop *Oct 2024*
Principal Speaker

- Presentation on published conference paper *Attention-fusion Model for Multi-Omics (AMMO) Data Integration in Lung Adenocarcinoma.* (Recording, Poster)

SELECTED PUBLICATIONS

[1] **W. Li**, A. Muneer, M. Waqas, X. Zhou, and J. Wu, "Attention-fusion model for multi-omics (ammo) data integration in lung adenocarcinoma," in *International Workshop on Computational Mathematics Modeling in Cancer Analysis*, pp. 52–60, Springer, 2024.

[2] M. B. Saad, Q. Al-Tashi, L. Hong, **W. Li**, D. Boiarsky, S. Li, M. Petranovic, C. C. Wu, B. Carter, T. Cascone, *et al.*, "1236 machine learning-based clinico-genomic prediction of benefits to add chemotherapy to immunotherapy in metastatic non-small cell lung cancer," 2024.

[3] **W. Li**, M. Kim, K. Zhang, H. Chen, X. Jiang, and A. Harmanci, "COLLAGENE enables privacy-aware federated and collaborative genomic data analysis," *Genome Biology*, vol. 24, no. 1, p. 204.

[4] **W. Li**, H. Chen, X. Jiang, and A. Harmanci, "Federated generalized linear mixed models for collaborative genome-wide association studies," *iScience*, vol. 26, no. 8, p. 107227.

[5] **W. Li**, H. Chen, X. Jiang, and A. Harmanci, "Fedgmmat: Federated generalized linear mixed model association tests," *PLoS computational biology*, vol. 20, no. 7, p. e1012142, 2024.

[6] **W. Li**, J. Tong, M. M. Anjum, N. Mohammed, Y. Chen, and X. Jiang, "Federated learning algorithms for generalized mixed-effects model (GLMM) on horizontally partitioned data from distributed sources," *BMC Medical Informatics and Decision Making*, vol. 22, no. 1, p. 269. Publisher: Springer.

[7] **W. Li***, J. Kim*, T. Bath, X. Jiang, and L. Ohno-Machado, "VERTIcal grid lOgistic regression with confidence intervals (VERTIGO-CI)," *AMIA Summits on Translational Science Proceedings*, vol. 2021, p. 355. Publisher: American Medical Informatics Association.

[8] Y. Huang, **W. Li**, F. Macheret, R. A. Gabriel, and L. Ohno-Machado, "A tutorial on calibration measurements and calibration models for clinical prediction models," *Journal of the American Medical Informatics Association*, vol. 27, no. 4, pp. 621–633. Publisher: Oxford University Press.